

Report on Physiological Datasets for Emotion Recognition

1. Introduction

The field of Affective Computing relies heavily on standardized datasets to benchmark algorithms for emotion recognition. The folders identified in your directory—DEAP and SEED_EEG—represent two of the most seminal datasets in this domain. While both aim to map human emotional states to physiological signals, they differ fundamentally in their experimental design, emotion models, and signal acquisition methodologies. This report details the technical specifications and research utility of each.

2. The DEAP Dataset

Full Title: A Database for Emotion Analysis using Physiological Signals **Origin:** A collaborative effort by Queen Mary University of London, University of Twente, University of Geneva, and EPFL.

Experimental Design and Stimuli

The DEAP dataset focuses on the analysis of spontaneous emotional responses elicited by multimedia content. The study involved 32 healthy participants, each of whom watched 40 one-minute excerpts of music videos. Music videos were selected as stimuli because they effectively induce strong emotional responses within a short timeframe. The experiment was designed to cover the varied space of the Valence-Arousal-Dominance model, ensuring a balanced representation of different emotional states.

Data Acquisition and Modalities

DEAP is a multimodal dataset, meaning it captures data from multiple physiological sources simultaneously. This allows researchers to study the correlation between central nervous system activity (brain waves) and the peripheral nervous system (body responses).

- **Central Nervous System (CNS):** Brain activity was recorded using a 32-channel Electroencephalogram (EEG) at a sampling rate of 512 Hz, which was subsequently downsampled to 128 Hz for easier distribution.
- **Peripheral Nervous System (PNS):** Eight additional channels recorded peripheral physiological signals, including Galvanic Skin Response (GSR) to measure arousal via sweat gland activity, skin temperature, blood volume pulse

(BVP), respiration rate, and electromyograms (EMG) of the Zygomaticus and Trapezius muscles to detect facial expressions and tension.

- **Behavioral Data:** For 22 of the 32 participants, frontal face video was also recorded, allowing for computer vision analysis of facial expressions.

Labeling and Emotion Model

Unlike datasets that classify emotions into discrete categories (e.g., Happy, Sad), DEAP utilizes a dimensional emotion model. After viewing each video, participants performed a self-assessment using the Self-Assessment Manikin (SAM) scale. They rated their emotional state on continuous scales of 1 to 9 for Valence (pleasantness), Arousal (excitation), Dominance (control), and Liking (preference). This makes DEAP particularly suitable for regression tasks and mapping emotions onto a continuous 2D or 3D space.

3. The SEED Dataset

Full Title: SJTU Emotion EEG Dataset Origin: Developed by the BCMI Laboratory at Shanghai Jiao Tong University.

Experimental Design and Stability

The defining characteristic of the SEED dataset is its focus on the temporal stability of EEG patterns. Neural signals are known to be non-stationary, meaning they change over time due to fatigue, electrode placement shifts, or mood changes. To address this, the SEED protocol required 15 participants to undergo the same experiment across three separate sessions, with each session spaced approximately one week apart. This unique design allows researchers to develop and test algorithms that are robust across different days, a critical requirement for real-world BCI applications.

Stimuli and Emotion Induction

Instead of music videos, SEED utilizes film clips to elicit emotions. In each session, participants watched 15 carefully selected Chinese film clips that were validated to elicit three specific discrete emotional states: Positive, Neutral, and Negative. This reliance on film clips is based on the premise that narrative visual content often induces more sustained and stable emotional patterns compared to music or abstract images.

Data Acquisition and Modalities

SEED focuses almost exclusively on high-density brain activity. It provides EEG recordings from 62 channels, offering a much higher spatial resolution than the 32-

channel setup used in DEAP. This high density allows for more precise source localization and the extraction of complex spatial features, such as Differential Entropy (DE) and Power Spectral Density (PSD), which are commonly used with this dataset. Some distributions of the dataset also include eye-tracking data to correlate gaze patterns with emotional states.

Labeling and Application

In contrast to DEAP's dimensional approach, SEED uses a discrete categorical model. The ground truth labels are simply Positive, Neutral, or Negative. This simplifies the machine learning task into a three-class classification problem. Consequently, SEED is frequently used to benchmark the performance of deep learning models, particularly in evaluating cross-subject (training on some subjects and testing on others) and cross-session (training on day 1 and testing on day 2) generalization capabilities.

4. Summary of Differences

The choice between these two datasets depends entirely on the research question. DEAP is the superior choice for multimodal fusion studies where the goal is to combine brain waves with heart rate or skin conductance to predict subtle shifts in arousal or valence. SEED, however, is the standard for pure EEG analysis when the goal is to build robust, stable classifiers that can function reliably over time or across different individuals, specifically distinguishing between positive and negative valence.